

DS lab8.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here x DS lab 9.c x *DSlab 7.c x DS lab8.c x

```
1  #include <stdio.h>
2  #define MAX 5
3  int pq[MAX];
4  int size = 0;
5  void enqueue(int value)
6  {
7      int i;
8      if(size == MAX)
9      {
10         printf("Priority Queue Overflow\n");
11         return;
12     }
13     for(i = size - 1; i >= 0 && pq[i] > value; i--)
14     {
15         pq[i + 1] = pq[i];
16     }
17     pq[i + 1] = value;
18     size++;
19     printf("%d inserted\n", value);
20 }
21 void dequeue()
22 {
23     int i;
24     if(size == 0)
25     {
26         printf("Priority Queue Underflow\n");
27         return;
28     }
29     printf("%d deleted\n", pq[0]);
30     for(i = 0; i < size - 1; i++)
31     {
32         pq[i] = pq[i + 1];
33     }
34     size--;
35 }
```

C:\Users\Sevanthi-v\Downloads\DS lab8.cC/C++Windows (CR+LF)WINDOWS-1252Line 1, Col 19, Pos 18InsertRead/Writedefault

DS lab8.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

Start here DS lab 9.c *DSlab 7.c DS lab8.c

```
31 {
32     pq[i] = pq[i + 1];
33 }
34 size--;
35 }
36 void display()
37 {
38     int i;
39     if(size == 0)
40     {
41         printf("Priority Queue is Empty\n");
42     }
43     else
44     {
45         printf("Priority Queue Elements:\n");
46         for(i = 0; i < size; i++)
47         {
48             printf("%d ", pq[i]);
49         }
50     }
51     printf("\n");
52 }
53 int main()
54 {
55     enqueue(30);
56     enqueue(10);
57     enqueue(50);
58     enqueue(20);
59     display();
60     dequeue();
61     display();
62     return 0;
63 }
64
65
```

C:\Users\Sevanthi-v\Downloads\DS lab8.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 1, Col 19, Pos 18 Insert Read/Write default

```
"C:\Users\Sevanthi-v\Downlo X + v
30 inserted
10 inserted
50 inserted
20 inserted
Priority Queue Elements:
10 20 30 50
10 deleted
Priority Queue Elements:
20 30 50

Process returned 0 (0x0) execution time : 0.108 s
Press any key to continue.
```